

Database System



Project Proposal

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Contents

- Problem Statement..... 2
- Problem Description 2
- Solution Approach..... 2
 - Rule-Based System 2
 - Web Based Management Portal 3
 - Tools and Technologies..... 3
 - Key Features 3
- Scope and Features 3
 - Main Functions..... 3
 - Scalability 4
 - User Features..... 4
 - Future Enhancements 4
- Conclusion..... 4

PROJECT PROPOSAL

Sports League Management System

PROBLEM STATEMENT

Managing a sports league manually is inefficient and error-prone. Administrators constantly struggle to maintain player records, schedule matches, track scores, and update standings in real time. A centralized digital system is needed to streamline operations and ensure data accuracy across all league activities.

PROBLEM DESCRIPTION

League management involves multiple interdependent entities. The key challenges include:

- ▶ Maintaining accurate records of teams, players, coaches, and referees.
- ▶ Scheduling matches across multiple venues and seasons without conflicts.
- ▶ Tracking live scores, standings, and individual player statistics.
- ▶ Recording player injuries and monitoring availability.
- ▶ Managing sponsors, tickets, fan registrations, and equipment.
- ▶ Generating performance reports and season-end awards.

SOLUTION APPROACH

The proposed system uses a two-layered approach combining a structured relational database with a web-based management portal.

Rule-Based System

The foundation is a normalized relational database built on the following principles:

- ▶ ER Diagram modeling all 20 entities and their relationships.
- ▶ Tables normalized to 3NF to eliminate data redundancy.
- ▶ Primary and foreign key constraints for referential integrity.
- ▶ Business rules enforced through SQL constraints and triggers.

- ▶ Core tables: Teams, Players, Matches, Scores, Standings, Injuries, Venues, Referees, Seasons, and more.

Web Based Management Portal

- ▶ A browser-accessible portal for managing all league data.
- ▶ Role-based access for administrators, coaches, referees, and fans.
- ▶ Real-time dashboard displaying standings, match schedules, and top scorers.
- ▶ Forms for data entry — player registration, match results, injury reports.
- ▶ Search and filter functionality across all league records.

Tools and Technologies

- ▶ MySQL — Relational database management and query execution.
- ▶ HTML / CSS / JavaScript — Web portal frontend.
- ▶ PHP — Backend server-side scripting.
- ▶ Draw.io — ER Diagram design and documentation.
- ▶ MS Word — Project documentation and reporting.

Key Features

- ▶ Centralized relational database for all league data.
- ▶ Automated standings calculation after each match result.
- ▶ Complete team, player, and coach profile management.
- ▶ Role-based web portal for all stakeholders.
- ▶ Advanced SQL reporting for statistics and season history.

SCOPE AND FEATURES

Main Functions

- ▶ Register and manage teams, players, coaches, and referees.

- ▶ Schedule and record match details, scores, and results.
- ▶ Track player statistics and injuries across seasons.
- ▶ Process ticket sales and fan registrations.

Scalability

- ▶ Supports multiple leagues and sports with minimal structural changes.
- ▶ Additional seasons and historical data added without redesign.
- ▶ Web portal expandable with mobile-responsive design.
- ▶ External API integration for live score feeds possible in future.

User Features

- ▶ Web portal accessible from any browser without installation.
- ▶ Live dashboard showing standings, upcoming matches, and top scorers.
- ▶ Search and filter for players, teams, and match history.
- ▶ Downloadable reports and season summaries.

Future Enhancements

- ▶ Mobile application for real-time league updates.
- ▶ Live score streaming via external sports data APIs.
- ▶ Machine learning based player performance predictions and lineup recommendations.
- ▶ Fan engagement features: polls, fantasy leagues, and voting.

CONCLUSION

This project provides a practical and scalable solution for sports league management. By combining a well-structured relational database with an interactive web portal, it eliminates manual inefficiencies and ensures reliable, real-time access to all league data. The flexible architecture supports future expansion, making it suitable for both academic evaluation and real-world deployment.